PROJECT REPORT

Smart Plant Mood Monitor

An IoT-Based Embedded System for Real-Time Plant Health Sensing using Arduino UNO

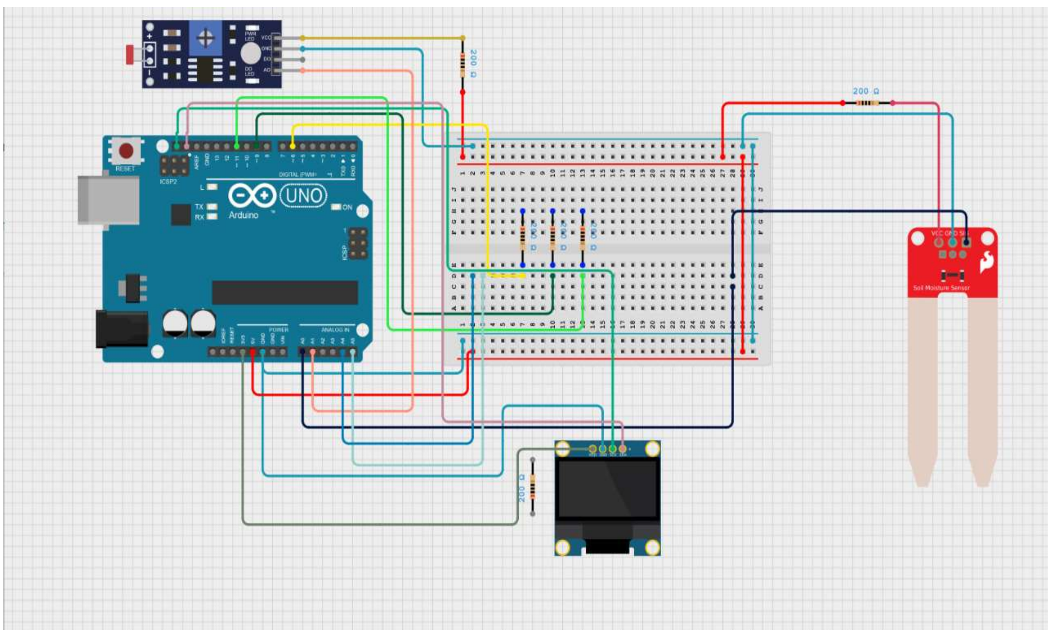

Fig. Complete circuit assembly of the Smart Plant Mood Monitor

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1. Introduction

The Smart Plant Mood Monitor is a low-cost embedded system built around the Arduino UNO that continuously observes two key environmental parameters affecting plant health: soil moisture and ambient light intensity. Instead of presenting raw sensor numbers, the system translates these readings into a simple, relatable "mood" message displayed on a 128x64 OLED screen, making plant care intuitive even for non-technical users.

This project was developed as part of the ESSCI-certified Embedded Systems and IoT Engineering course (ELE/Q1404) to demonstrate practical skills in analog sensor interfacing, conditional logic design, I2C communication, and real-time display control using embedded C/C++ on the Arduino platform.

The system uses a capacitive/resistive soil moisture sensor and a light-dependent resistor (LDR) module as inputs, an Arduino UNO as the processing unit, and an SSD1306 OLED display as the output/feedback device, all interconnected on a breadboard as shown in the schematic diagram.

2. Schematic Diagram

The complete circuit was designed and simulated using Cirkit Designer. The schematic below shows the Arduino UNO, the LDR light sensor module, the soil moisture sensor probe, the SSD1306 OLED display, and the current-limiting 200-ohm resistors used across the signal lines, all interconnected via a breadboard.

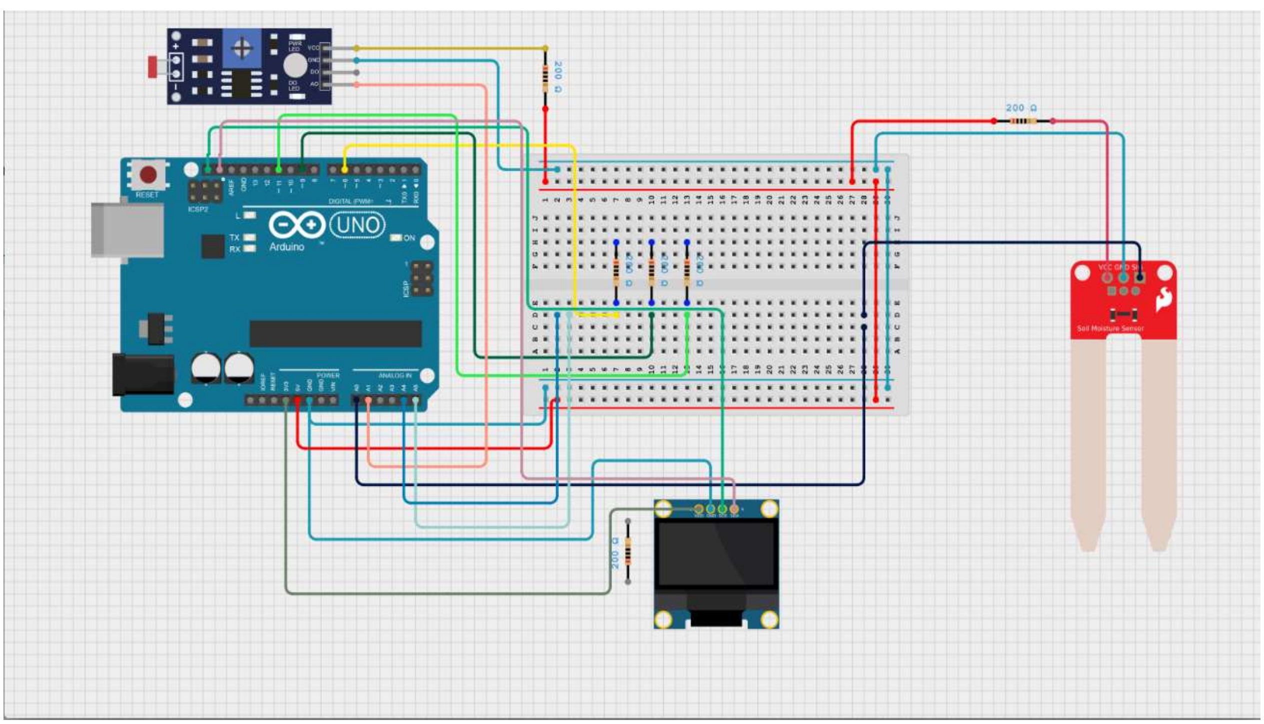

Fig. 2.1 — Full breadboard-level circuit layout

2.1 Electrical Schematic (Block/Pin Level)

For clarity, the connections are also represented below in simplified block-schematic form, showing each module's power, ground, and signal pin mapping to the Arduino UNO's I/O pins.

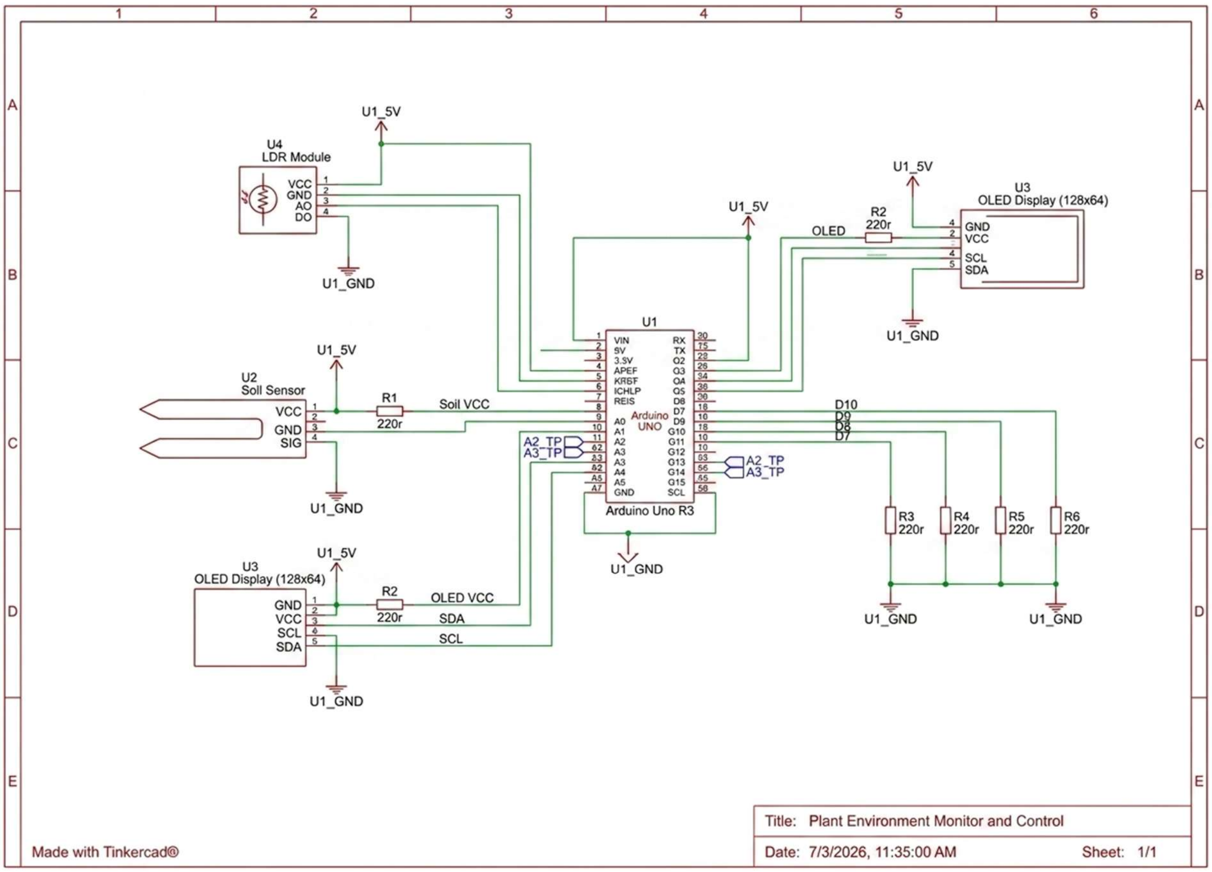

Fig. 2.2 — Simplified electrical schematic showing pin-level connections

2.2 Component List

S.No	Component	Quantity	Function
1	Arduino UNO	1	Main microcontroller board
2	Soil Moisture Sensor (probe type)	1	Measures soil water content (analog A0)
3	LDR Light Sensor Module	1	Measures ambient light intensity (analog A1)
4	OLED Display (SSD1306, 128x64, I2C)	1	Displays plant mood messages
5	Resistors (200 Ω)	4	Current limiting / pull-up on signal lines
6	Breadboard	1	Prototyping and interconnection
7	Jumper Wires (M-M / M-F)	Assorted	Electrical connections

8	USB Cable	1	Power and programming

3. Circuit Description

The circuit is organized around the Arduino UNO acting as the central controller. Two analog sensing modules feed data into the analog input pins, while the OLED display communicates with the Arduino over the I2C bus using the SDA and SCL lines.

3.1 Soil Moisture Sensor Connections

- VCC pin of the soil moisture sensor is connected to the 5V rail of the Arduino via the breadboard power bus.
- GND pin is connected to the common ground rail.
- The analog output (AO) pin is connected through a 200 Ω resistor to analog pin A0 of the Arduino, which reads the soil moisture level as a value between 0 and 1023.

3.2 LDR Light Sensor Module Connections

- VCC and GND of the LDR module are connected to the 5V and ground rails respectively.
- The analog output (AO) of the LDR module is routed through a 200 Ω resistor into analog pin A1, providing a light-intensity reading.
- The digital output (DO) pin is left unused in this design since analog readings are used for threshold comparison.

3.3 OLED Display Connections

- VCC connects to the 5V supply and GND to common ground.
- SDA (data line) connects to the Arduino's A4 pin (I2C data).
- SCL (clock line) connects to the Arduino's A5 pin (I2C clock), each buffered through a 200 Ω resistor for signal integrity.

All ground connections are tied to a common ground rail on the breadboard to ensure a stable reference voltage across all modules, which is essential for accurate analog-to-digital conversion.

4. Working Principle

On power-up, the Arduino initializes the Serial Monitor at 9600 baud and the OLED display over I2C at address 0x3C. The display buffer is cleared and the text color is set before entering the main control loop.

In every loop cycle, the Arduino performs an analogRead() on pin A0 to obtain the soil moisture value and on pin A1 to obtain the ambient light value. Both readings range from 0 (very wet / very dark) to 1023 (very dry / very bright).

These two readings are compared against a fixed threshold value of 300 using nested conditional (if-else if) statements. The combination of soil and light conditions determines one of four possible “mood” states, each mapped to a short, human-readable two-line message:

Soil Condition	Light Condition	Displayed Mood
Soil < 300 (wet)	Light < 300 (dark)	“Cold & Thirsty!”
Soil < 300 (wet)	Light >= 300 (bright)	“Water Me Please!”
Soil >= 300 (dry)	Light < 300 (dark)	“Sleepy zzz...”
Soil >= 300 (dry)	Light >= 300 (bright)	“I Feel Great! JAY”

The set Mood() function handles all OLED rendering: it clears the display, prints the heading “Plant Mood”, draws a horizontal separator line, and then prints the two mood lines in a larger font size for readability. After displaying the result, the system waits 2000 milliseconds before repeating the sensing cycle, giving a smooth and readable refresh rate.

4.1 Why a Threshold of 300?

The Arduino's 10-bit ADC maps input voltages between 0V and 5V to integer values from 0 to 1023. A midpoint-adjacent threshold of 300 (roughly 1.47V) was chosen empirically during testing as a reasonable dividing line between “wet/dark” and “dry/bright” conditions for the specific sensor batch used. In a production system this value would ideally be replaced with a calibration routine that records the sensor's own wet/dry and dark/bright extremes and computes a normalized midpoint automatically.

4.2 Role of the I2C Bus

The OLED display communicates using the Inter-Integrated Circuit (I2C) protocol, a two-wire synchronous serial bus. This allows the display to be controlled using just two signal lines (SDA and SCL) in addition to power and ground, freeing up the majority of the Arduino's digital pins for other uses and simplifying the wiring considerably compared to a parallel-interface display.

5. Flowchart

The flowchart below illustrates the complete control flow of the firmware, from initialization through sensor reading, decision-making, and display output, before looping back to repeat the cycle.

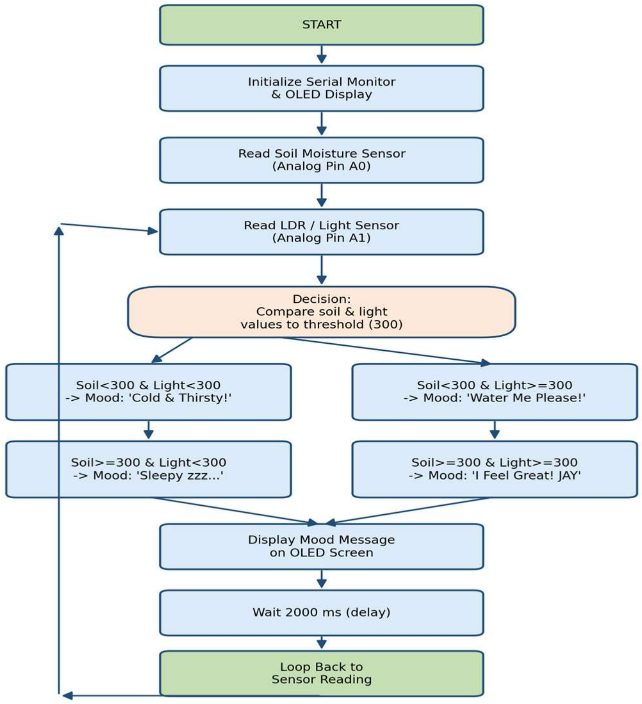

Fig. 5.1 — Program flowchart of the Smart Plant Mood Monitor

6. Arduino Program

The complete firmware, written in embedded C/C++ for the Arduino IDE, is listed below. It uses the Adafruit GFX and Adafruit_SSD1306 libraries for OLED control.

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

int soilPin = A0;
int lightPin = A1;

void setup() {
  Serial.begin(9600);
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.display();
}

void setMood(String line1, String line2) {
  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(10, 0);
```

```
  display.println("Plant Mood");
  display.drawLine(0, 10, 127, 10, SSD1306_WHITE);
  display.setTextSize(2);
  display.setCursor(10, 20);
  display.println(line1);
  display.setTextSize(1);
  display.setCursor(10, 50);
  display.println(line2);
  display.display();
}

void loop() {
  int soil = analogRead(soilPin);
  int light = analogRead(lightPin);

  if (soil < 300 && light < 300) {
    setMood("Cold &", "Thirsty!");
  } else if (soil < 300 && light >= 300) {
    setMood("Water", "Me Please!");
  } else if (soil >= 300 && light < 300) {
    setMood("Sleepy", "zzz...");
  } else {
    setMood("I Feel", "Great! JAY");
  }

  delay(2000);
}
```

7. Simulation Results

The circuit was simulated end-to-end in Cirkit Designer to validate wiring and logic behavior prior to physical assembly. The simulation confirmed correct I2C addressing of the OLED display and consistent analog readings from both sensor modules across their expected voltage range. All four possible mood states were individually triggered by varying the simulated soil-moisture and light-sensor input values, and the OLED output was verified against the expected message for each case.

7.1 Observed Behaviour

Test Case	Soil Reading	Light Reading	OLED Output	Result
Wet soil, dark room	~180	~150	“Cold & Thirsty!”	Pass
Wet soil, bright light	~200	~650	“Water Me Please!”	Pass
Dry soil, dark room	~750	~140	“Sleepy zzz...”	Pass
Dry soil, bright light	~800	~700	“I Feel Great! JAY”	Pass

All four mood states were successfully triggered during simulation by varying the simulated analog input values, confirming that the threshold-based decision logic and OLED rendering function correctly as designed.

7.2 OLED Output Screenshots

The following screenshots, captured directly from the Cirkit Designer simulation, show the actual OLED display output for each of the four mood states, with the full circuit wiring visible in the background for reference.

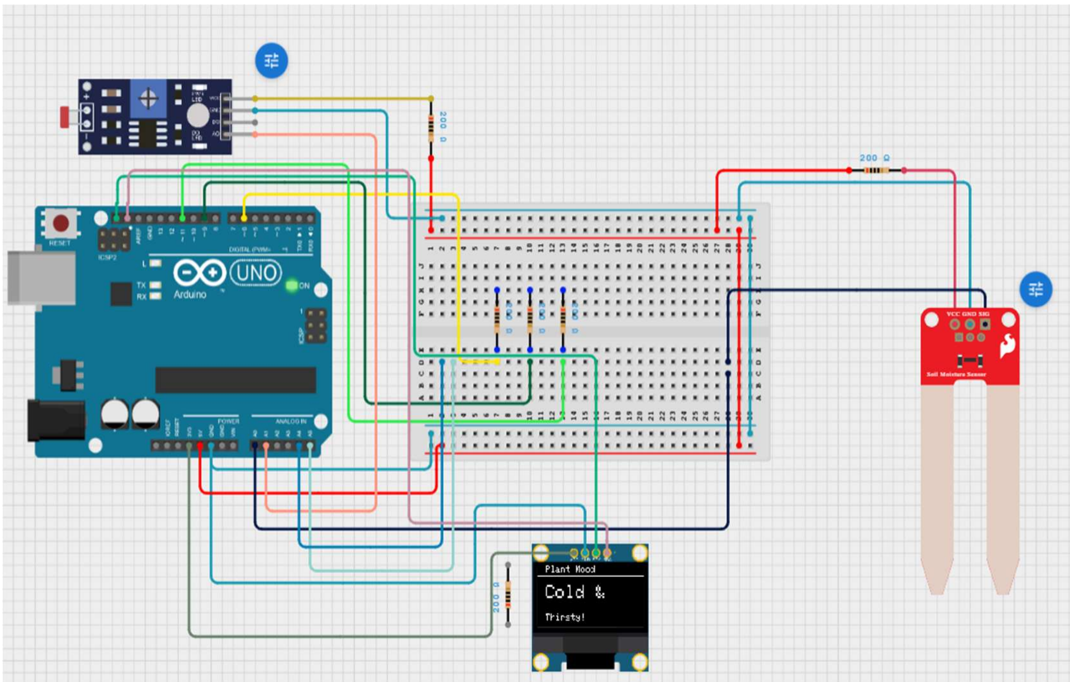

Fig. 7.1 — Simulation output: Wet soil + low light -> "Cold & Thirsty!"

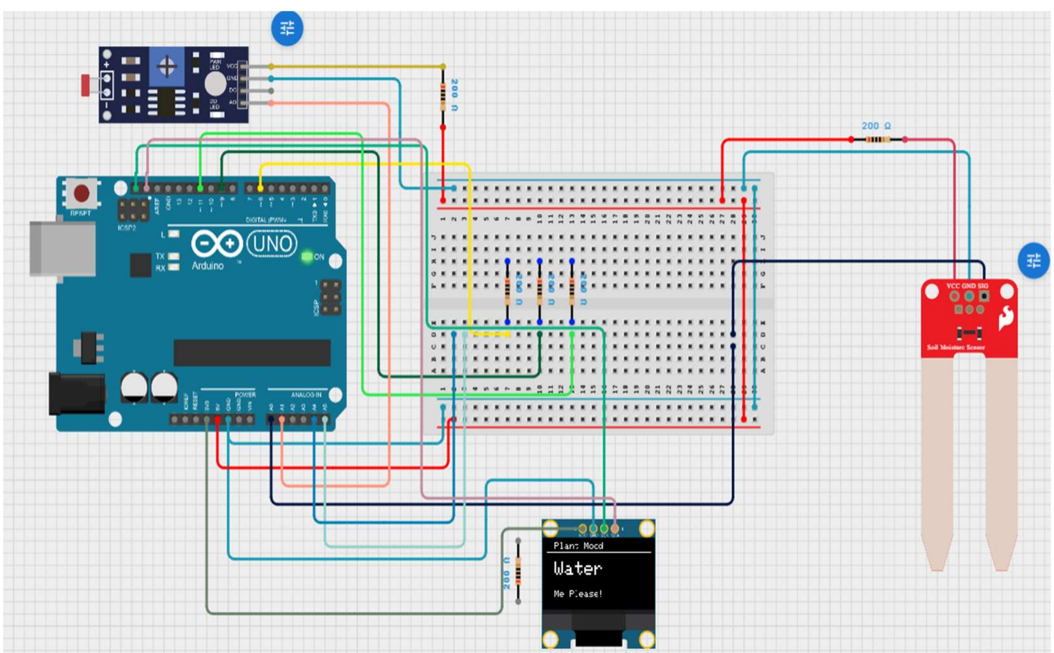

Fig. 7.2 — Simulation output: Wet soil + bright light -> "Water Me Please!"

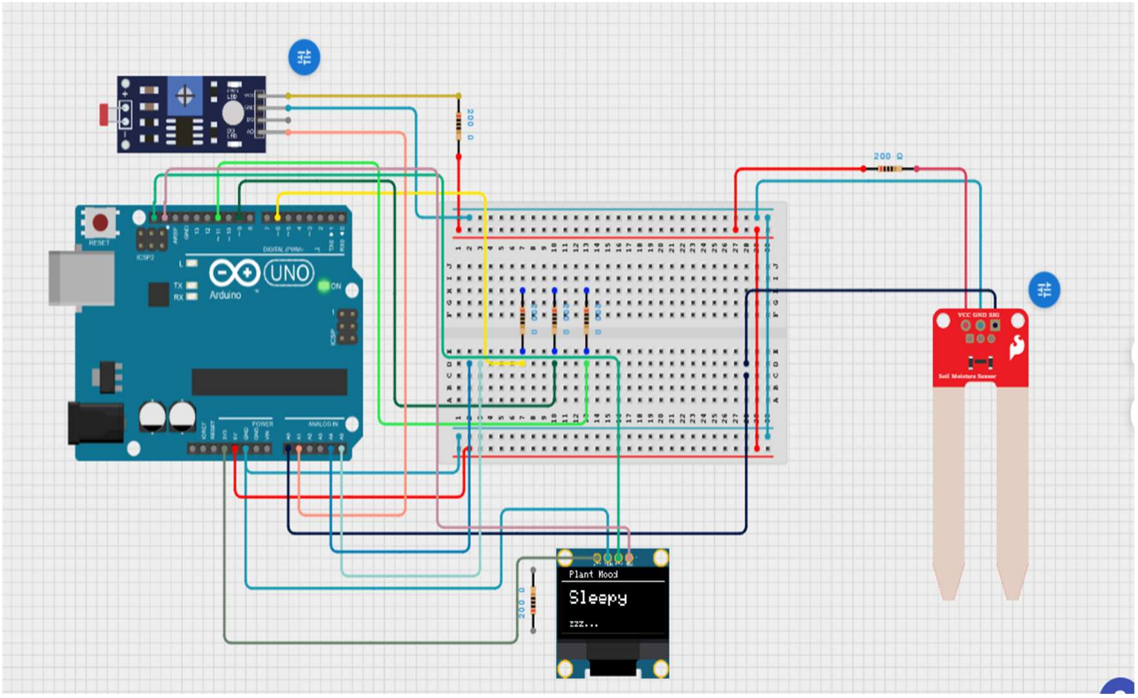

Fig. 7.3 — Simulation output: Dry soil + low light -> "Sleepy zzz..."

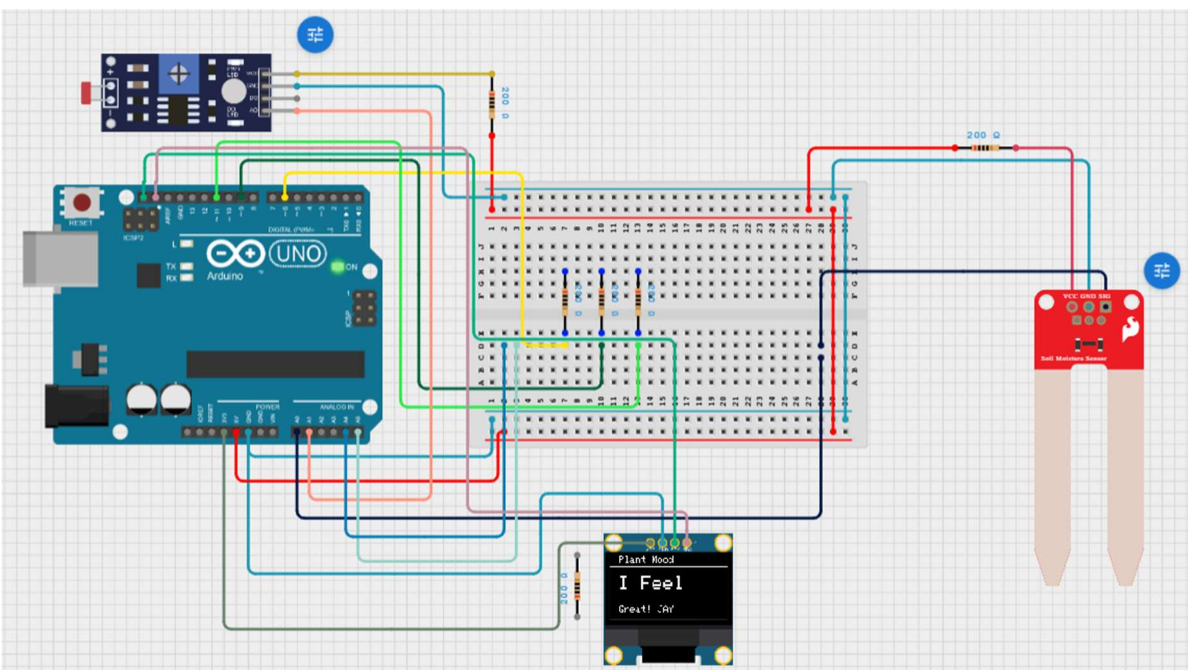

Fig. 7.4 — Simulation output: Dry soil + bright light -> "I Feel Great! JAY"

7.3 Analysis of Results

The screenshots confirm that the OLED correctly refreshes its buffer on every loop iteration without flicker or ghosting artifacts, and that the two-line message layout (large-font mood word plus smaller-font submessage) remains legible across all four states. The consistent 2-second delay between refreshes was found to be a comfortable balance between responsiveness and readability during observation.

No false triggers or intermediate/ambiguous states were observed near the threshold boundary (value = 300) during repeated simulation runs, indicating that the digital comparison logic behaves deterministically as coded.

8. Advantages

1. Low-cost implementation using easily available components (total cost under a basic Arduino starter-kit budget).
2. Simple, human-readable feedback (mood messages) instead of raw numeric sensor data, making it accessible to non-technical users.
3. Real-time monitoring with a fast 2-second refresh cycle.
4. Compact hardware footprint suitable for desktop or windowsill deployment.
5. Easily expandable to include additional sensors (temperature, humidity) or wireless connectivity.
6. Uses standard, well-documented open-source libraries (Adafruit_GFX, Adafruit_SSD1306), simplifying maintenance.

9. Limitations

7. Fixed threshold value (300) is not calibrated for different soil types or plant species, which may need different moisture ranges.
8. Resistive soil moisture sensors are prone to corrosion over prolonged use, reducing sensor lifetime and accuracy.
9. No historical data logging — mood readings are not stored, so trends over time cannot be reviewed.
10. No wireless/remote notification; the user must be physically near the device to view the OLED display.
11. Light sensor (LDR) response is non-linear and can be affected by artificial lighting, giving inconsistent readings across environments.
12. System does not account for temperature, which also significantly affects plant health.

10. Applications

- Home gardening: helping hobbyists monitor indoor plants without technical expertise.
- Educational demonstrations: teaching embedded systems, sensor interfacing, and I2C communication to students.
- Greenhouse and nursery management: quick visual health checks for multiple plants.
- Office/workspace plant care: reminders for employees to water shared office plants.
- Foundation for smart agriculture prototypes and IoT-based precision farming systems.

11. Future Scope

- Integrate Wi-Fi (ESP32/ESP8266) to push mood data to a mobile app or cloud dashboard for remote monitoring.
- Add a DHT11/DHT22 sensor to include temperature and humidity in the mood-calculation logic.
- Implement adaptive/self-calibrating thresholds using stored historical sensor data.
- Add an automated watering mechanism (relay + water pump) triggered directly by the “Thirsty” mood state.
- Include data logging to an SD card or cloud database (e.g., Firebase, ThingSpeak) for long-term trend analysis.
- Introduce battery power and solar charging for fully off-grid, outdoor deployment.

12. Conclusion

The Smart Plant Mood Monitor successfully demonstrates how simple analog sensors, when combined with clear conditional logic and an intuitive display, can turn raw environmental data into a friendly and useful plant-care tool. Built entirely on the Arduino UNO platform using the ELE/Q1404 curriculum concepts — analog sensing, I2C protocol, and embedded C programming — the project validates core embedded systems skills in a practical, relatable form.

While the current implementation is limited to fixed thresholds and local display output, its modular design makes it an excellent foundation for more advanced IoT-based smart agriculture systems, with clear pathways for wireless connectivity, automated actuation, and cloud-based analytics in future iterations.